

Hands-only CPR for sudden cardiac arrest

Sudden cardiac arrest is when the heart suddenly stops pumping blood. The person collapses, stops responding, and stops breathing normally (or only gasps). Without blood flow to the brain, every minute counts. A bystander who starts CPR right away can double or triple someone's chance of survival.

CALL EMERGENCY SERVICES FIRST

Collapsed and not breathing normally = emergency

Call your local emergency number (911 in the US/Canada, 112 in much of Europe, 999 in the UK) immediately, or tell a specific person to call and to find an AED. Put the phone on speaker so the dispatcher can guide you. Do not wait to be sure — if someone is unresponsive and not breathing normally, start CPR.

Three quick checks

1. **Check for a response.** Tap their shoulders firmly and shout, “Are you OK?” If they do not respond, they need help now.
 2. **Check breathing.** Look at the chest for no more than about 10 seconds. If they are not breathing, or are only gasping, struggling, or making occasional noisy breaths, treat it as cardiac arrest.
 3. **Call and get an AED.** Call emergency services (speaker phone) and send someone for the nearest automated external defibrillator (AED). If you are alone with a phone, call first, then start CPR.
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Gasping — slow, noisy, irregular breaths in the first minutes — is **not** normal breathing. It is a sign of cardiac arrest. Start CPR.

How to do hands-only CPR (adults & teens)

PUSH HARD, PUSH FAST, DON'T STOP

1. **Kneel beside them** on a firm, flat surface. Place the heel of one hand on the **centre of the chest** (on the breastbone, between the nipples). Put your other hand on top and interlock your fingers.
2. **Push down hard and fast.** Keep your arms straight and your shoulders directly over your hands, and press straight down at least **5 cm (2 inches)** deep — about one-third the depth of the chest — but not more than about 6 cm (2.4 inches).
3. **Let the chest come all the way back up** between pushes (full recoil) without lifting your hands off. This lets the heart refill.
4. **Keep a fast, steady rate of 100–120 pushes per minute** — about two per second. The beat of many well-known upbeat songs works as a guide.
5. **Do not stop** except to use an AED or to swap with someone. Switch with another bystander about every 2 minutes if you can, to keep pushes strong — good compressions are tiring.

For an untrained rescuer helping an adult or teen, **hands-only CPR** — continuous chest compressions with no rescue breaths — is recommended and effective. If you are trained and willing, give 30 compressions then 2 rescue breaths, but never let giving breaths delay or interrupt good chest compressions for long.

Use an AED as soon as one arrives

AEDS ARE MADE FOR UNTRAINED BYSTANDERS

An AED is a small device, often found in airports, schools, gyms, offices, and shopping centres. **Turn it on and follow the spoken instructions** — it tells you exactly what to do. Stick the pads on the person's bare chest as shown in the pictures on the pads. Keep doing chest compressions while the pads go on. When the AED says “analysing,” make sure no one is touching the person. If it advises a shock, make sure everyone is clear, then press the button. Right after any shock or any “no shock” message, **resume compressions immediately** and keep following the AED.

When to stop

- Emergency responders or another trained person takes over.
- The person clearly wakes up, moves, opens their eyes, and starts to breathe normally. If so, roll them onto their side (recovery position) and keep watching their breathing.
- An AED instructs you to stop while it analyses or delivers a shock (resume immediately afterward).
- You are too exhausted to continue and no one can take over.

Children, babies, drowning, and overdose

BREATHS MATTER MORE HERE

In babies and children, and when collapse is caused by drowning, choking, or a drug overdose, lack of oxygen is usually the cause, so **rescue breaths are especially valuable**. If you are trained, give breaths with compressions (commonly 30 compressions to 2 breaths; trained child/infant responders often use 15:2 with two rescuers). For a baby, use two fingers in the centre of the chest and push about 4 cm (1.5 inches) deep. If you are not trained, the dispatcher can guide you — and doing chest compressions alone is far better than doing nothing.

Good to know

- **You will not "do it wrong" by trying.** The biggest mistake is doing nothing. A person in cardiac arrest will not survive without help.
- **Cracked ribs can happen** and are not a reason to stop — effective compressions save lives.
- **Don't check for a pulse** if you are untrained — it wastes time and is easy to get wrong. Go by responsiveness and breathing.
- A heart attack (a blocked artery, often with chest pain) is different from cardiac arrest, but a heart attack can lead to cardiac arrest — so if someone with chest pain collapses and stops breathing normally, start CPR.
- **Learn CPR in person** if you can. Short, low-cost or free classes are widely available and build the confidence to act.

Sources

- American Heart Association (AHA) — *Hands-Only CPR* and CPR & ECC guidelines (compression rate 100–120/min, depth at least 2 in / 5 cm, full recoil, minimise interruptions). heart.org, cpr.heart.org
- American Red Cross — CPR and AED first-aid guidance for adults, children, and infants. redcross.org
- Resuscitation Council UK — adult and paediatric basic life support guidelines. resus.org.uk
- European Resuscitation Council (ERC) — basic life support and AED guidelines. erc.edu

Steps reflect widely taught basic-life-support sequences for lay responders. Training bodies differ in small details — follow your local first-aid training and your emergency dispatcher's instructions.

This guide is general information, not medical advice, and is not a substitute for hands-on CPR and AED training, which is widely available and strongly recommended. In an emergency, always call your local emergency number. Reuse and reprint freely.